

Web Programming – Week 3 – Working Questions

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Question 1 – Conceptual (Understanding Flexbox)

Answer the following questions in simple and clear sentences:

1. What is Flexbox in CSS?
2. Why do we use Flexbox in web design?
3. What problems does Flexbox solve compared to older methods like float?
4. What is the difference between:
 - A flex container
 - A flex item
5. Give one example where Flexbox is useful (for example: navigation bar, card layout, etc.).

Question 2 – Modifying the Web Page Using Flexbox

You are given the following HTML and CSS files.

Important: You are NOT allowed to modify the HTML file. You must modify only the CSS file.

Initial HTML (index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Flexbox Exercise</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<nav>
  <div class="logo">LOGO</div>
  <div class="menu">Home</div>
  <div class="menu">About</div>
  <div class="menu">Contact</div>
</nav>

<section class="main">
  <div class="box box1">Box 1</div>
  <div class="box box2">Box 2</div>
  <div class="box box3">Box 3</div>
</section>

<footer>
  Footer Area
</footer>

</body>
</html>
```

Initial CSS (style.css)

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
}

nav {
  background-color: lightgray;
  padding: 10px;
}

.main {
  background-color: #f4f4f4;
}

.box {
  padding: 20px;
  background-color: lightblue;
  margin: 10px;
}

footer {
  background-color: gray;
  color: white;
  text-align: center;
}
```

Your Tasks

Task 1 – Navigation Bar

Modify the navigation bar using Flexbox:

- Use `display: flex`.
- Place the items in one row.
- Use `justify-content` to distribute space between items.
- Use `align-items` to center items vertically.
- Set the height of the navigation bar to `10vh`.

Task 2 – Main Section Layout

Modify the `.main` section:

- Use Flexbox.
- Display the three boxes in one row.
- Use:

- `flex-grow`
 - `flex-shrink`
 - `flex-basis`
- Make Box 1 grow more than the others.
- Make Box 3 shrink more than the others.

Resize your browser and observe what happens to the boxes.

Task 3 – Viewport and Percentage Units

- Set the height of:
 - `.main` to `70vh`
 - `footer` to `20vh`
- Use percentage widths (for example `30%`, `40%`, `30%`) for the boxes.
- Use percentage margin in at least one place.

Expected Layout (Example Sketch)

	LOGO	Home	About	Contact		

	Box 1		Box 2		Box 3	

	Footer					

The final layout should clearly show:

- Proper use of Flexbox
- Different growing and shrinking behavior
- Use of viewport units (`vh`)
- Use of percentage units (`%`)